Tutorial - 10

S.NO	Feature 1 (Age)	Feature 2 (Income)	Feature 3 (credit score)	Target (loan approved)
S_1	25	40	650	yes
S_2	30	60	700	yes
S_3	45	35	600	No
S_4	50	30	580	No
S_5	28	55	720	yes

Target encoding

yes = -1

NO = 1

Questions-

1) Assign initial weights to all 5 samples.
2) construct a decision stump using feature 1 (Age) to classify the samples.
3) Identify misclassified samples and compute the weighted error rate.
4) calculate the classifier weight
5) update the sample weights for the next iteratio

Sol: AdaBoosting (Adaptive Boosting)

1) Initialize weights for all.
2) Check how many correct classification & in correct classifications for each attribute.
3) Find weighted error $\rightarrow \epsilon_\alpha$

$$\epsilon_\alpha = \begin{pmatrix} \text{no. of} \\ \text{incorrect ones} \end{pmatrix} \times \begin{pmatrix} \text{weights of} \\ \text{each} \end{pmatrix}$$

4) $\alpha x = \frac{1}{2} \log_e \left(\frac{1-e^x}{ex} \right)$ → performance of Decision stamp.

5) calculate new weights for each

$W_{new} = W_{old} \times e^{\alpha}$ → Incorrect classification

$W_{new} = W_{old} \times e^{-\alpha_1}$ → correct classification.

6) Normalize weights → $\frac{W_{new}}{(\text{Total})}$ → By adding all weights

7) For all attributes follow same.

8) final predicted values

$P = \alpha_1 h_1(x) + \alpha_2 h_2(x) + \cdots \alpha_n h_n(x)$

$h_1(x)$ → yes → take (1)

$h_1(x)$ → NO → take (0)

if $P > 0$ → yes } After calculating
$P < 0$ → NO } After Average

Adaptive Boosting is an ensemble machine learning that combines multiple weak Learners (called decision stamps). To form a strong predictive model.

Step: 1- Given data,

Sample	Age	Loan approved
S_1	25	yes +1
S_2	30	yes +1
S_3	45	NO −1
S_4	50	NO −1
S_5	28	yes +1

Assign initial weights

Total samples = 5

weight $(w_1) = \frac{1}{5} = 0.2$

$S_1 = S_2 = S_3 = S_4 = S_5 = 0.2.$

Step: 2- Construct Decision stamp
(Feature Age)

Sort values

Sample	Age	Loan Approved
S_1	25	+1
S_5	28	+1
S_2	30	+1
S_3	45	-1
S_4	50	-1

Chase Threshold :- Age $\leq 28 \rightarrow$ yes

Age $\geq 28 \rightarrow$ NO

Step: 3- Misclassified samples are : S_2

where Table is:-

Sample	Age	Loan approved	predicted
S_1	25	yes (+1)	+1
S_5	28	yes (+1)	+1
S_2	30	yes (+1)	-1
S_3	45	NO (-1)	-1
S_4	50	NO (-1)	-1

Step: 4- weighted error

$e = 1 \times (0.2) = 0.2$

Step:5 - calculate α (Needed for weight update)

$$\alpha = \frac{1}{2} \log\left(\frac{1-ex}{ex}\right) \rightarrow \text{feature}$$

$$= \frac{1}{2} \log\left(\frac{1-(0.2)}{0.2}\right)$$

$$\alpha = 0.693$$

Step:6 - update weights

for correctly classified $= e^{-\alpha} = e^{-0.693} = 0.5$

wrongly classified $= e^{\alpha} = e^{0.693} = 2$

Previous weight (old)

$$S_1 = S_2 = S_3 = S_4 = S_5 = 0.2$$

New weight →

$$S_1 = S_3 = S_4 = S_5 = 0.5 \Rightarrow 0.2 \times 0.5 = 0.1$$

$$S_2 = 2 \Rightarrow 0.2 \times 2 = 0.4$$

$S_1 = 0.1$, $S_2 = 0.1$, $S_3 = 0.1$, $S_4 = 0.1$, $S_5 = 0.1$

Total weights $= 0.1 + 0.4 + 0.1 + 0.1 + 0.1 = 0.8$

⇒ Normalize weights -

Sample	Final weights
S_1	$0.1/0.8 = 0.125$
S_2	$0.4/0.8 = 0.5$
S_3	$0.1/0.8 = 0.125$
S_4	$0.1/0.8 = 0.125$
S_5	$0.1/0.8 = 0.125$